

NAME: _____

PERIOD: _____

Two Payloads, One Regression Line

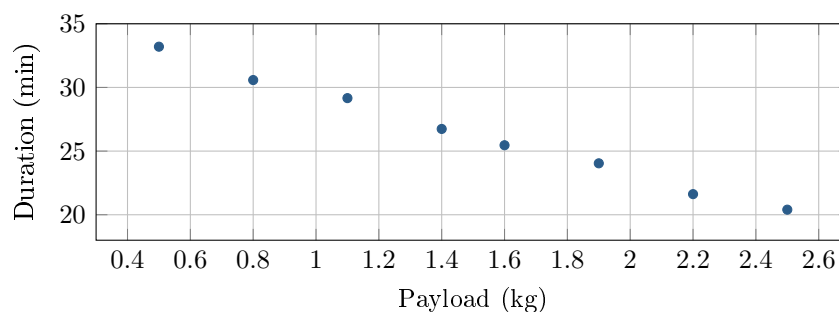
AP Statistics · Unit 5 · Topic 5.3

[STAR] THE PROBLEM

Suppose eight drone tests with payloads from 0.5 to 2.5 kg produced the shown data and the model $\hat{y} = 36.0 - 6.4x$, where x is payload and $\hat{y}$ is predicted flight duration in minutes. A proposed mission carries 3.5 kg and requires 12 minutes of flight.

Ravi: Approve it: the model predicts more than the required 12 minutes, and the fitted data are close to a line.

Mina: Before deciding, compare the support for predictions at 1.8 and 3.5 kg and identify evidence the safety decision still needs.



FIRST TAKE — one sentence before you work the parts

Would you approve now, delay approval, or approve conditionally? Cite one number or plot feature.

[BOOK] USING A REGRESSION MODEL (keep on the page)

A linear model $\hat{y} = a + bx$ predicts response y from explanatory variable x . Its slope is the predicted change in y for a one-unit increase in x ; substitution gives a prediction, not an exact outcome. Interpolation uses an x inside the observed range. Extrapolation uses an x outside that range and becomes less reliable farther from the data because the fitted pattern may not continue. A plausible output does not create evidence beyond the observations used to fit the model.

Work (a)–(c).

DOK 1 (a) Name the explanatory and response variables, with units.

DOK 2 (b) Predict flight duration at 1.8 and 3.5 kg. How much does the 3.5-kg prediction exceed the required 12 minutes?

DOK 3 (c) Would you approve this mission now? Compare the support for the two predictions using the tested payload range and the extra time from (b). Explain what evidence you would want before relying on the 3.5-kg prediction. Write 3–4 sentences.

Frame: At 1.8 kg the model gives _____, supported by _____; at 3.5 kg it gives _____, supported by _____.

Frame: For this mission I would _____ because _____; before relying on the model, collect _____.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored E / P / I.